



Investigation into Wake Interactions of Wind Lenses at Close Proximities

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Abstract

The aerodynamic interactions between two lens profiles that lead to a net increment in power of 10% from a wind lens turbine (WLT) is quantified. The lens wake interactions as a function of normalized distance (x/d) is investigated using Particle Image Velocimetry (PIV) as well as hotwire anemometry for three pairs of lens profiles with different geometries. All experiments were conducted at the University of Dayton's Low Speed Wind Tunnel (UD-LSWT). PIV and hotwire results indicate that there is an inherent bias in the wake properties at the lens spacings of $0.2 x/d$ and below, regardless of the lens geometry. This bias manifests as a larger momentum wake deficit and a thicker wake for one of the lenses when compared to its counterpart. The merged wake location is skewed to a single lens in the lens-pair depending on the extent of cancellation of inboard vorticity magnitude between the lens pairs. Around 0.1 to $0.2 x/d$, the individual wakes from each lens pair start to merge as one and coincidentally, it is at this spacing the net power output reaches its maximum. The increment in power output is due to the removal of high frequency vortex shedding in the merged wake.

I. Nomenclature

C_d	= Drag coefficient
C_P	= Power coefficient
h	= Height of the flange, m
L_D	= Diffuser length, m
L_N	= Nozzle length, m
P	= Electric Power, W
ρ	= Density, kg/m^3
R	= Rotor Radius, m
R_D	= Diffuser height, m
R_N	= Nozzle height, m
U_∞	= Freestream velocity, m/s

II. Introduction

With the need for clean and sustainable energy production at an all-time high, utilizing readily available resources like wind presents a major solution in attaining this need. Based on current projections, it is estimated that wind will provide 35% of the nation's energy needs by 2050 [1]. As the wind power is proportional to the cube of wind speed, even slight increments in freestream velocity can result in large increase in the power output. Recent methods have seen an increase power output of turbine systems by increasing rotor size. However, large turbine blades have had negative effects both environmentally and logistically such as manufacturing, transportation, installation, and maintenance and thereby creating problems in terms of availability and large-scale production. Additionally, the sheer size of these turbines threatens local wildlife, especially birds. However, by utilizing a more compact, higher efficiency turbine arranged in a grid pattern, some of these negative effects could be overcome without sacrificing power output and efficiency significantly. A great contender to large turbines is wind lens turbines (WLTs), which

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can increase power production by a factor of 2-5 when compared to a conventional turbine of similar size [2], [3], [4], [5].

A cross-sectional view of this WLT is displayed in **Fig. 1**. Utilization of the lens, or shroud around a turbine increases the overall efficiency of the system as the lens accelerates the flow passing through the turbine. The performance can further be increased by arranging the WLTs in grid configurations, thereby minimizing the footprint and consequently limiting the associated habitat destruction and logistical costs. The increased power output is attributed to the positive aerodynamic interactions amongst the lenses, yet this interaction largely remains unexplored. The objective of the current research is to study the wake interactions around the lenses under different lens profiles and spacing. The primary aerodynamic influence of a turbine on the other is predominantly due to the recirculation regions that form behind the lens (**Fig. 1**). These recirculation regions produce vortex shedding in the wake of the lens which interact to a certain extent depending on the size and proximity of each WLT in the array. From a fluid dynamic standpoint, it is not clear what the interaction does to the strength of the vortices behind the lens and how it affects the overall pressure in the flow field. It is also unclear what happens to these interactions as a function of time because the lenses produce vortices periodically with a certain shedding frequency. The work presented in this paper focuses on elucidating these interactions both spatially and temporally.

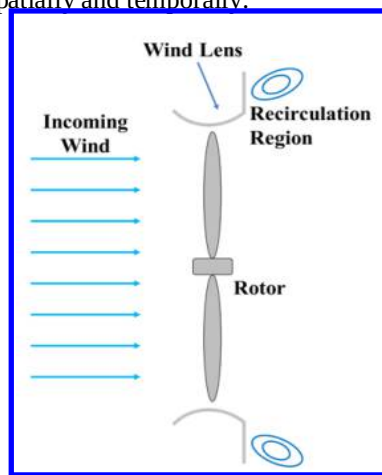


Figure 1: Schematic of Wind Lens Turbine

Results from similar studies involving side-by-side cylinders as well as flat plates were also referenced as the aerodynamic interactions between such bluff bodies provide insight into the unique profile of the lens shroud. Investigations conducted by Zhou, Zang, & Yiu [6] on side-by-side cylinders indicates the early termination of one downstream wake, due to a significant difference in the vortex strengths between the two. On the other hand, Teimourian, Yazdi, & Hacisevki [7], investigated two flat plates side-by-side and focused on the observable Strouhal numbers associated with each wake region, where small-scale vortices shed at a higher frequency than the large-scale vortex street, indicated by a double peak in the FFT spectral analysis. Both observations suggest that two separate vortex flow regimes form behind the bluff bodies, leading to the dominance of one over the other, and thus an amalgamation of the two at some distance downstream. Based on the observations of these, and other similar studies, the current research was aimed at identifying both the momentum deficit downstream the lens profiles, along with the associated fluctuating flow characteristics present at close proximities. By analyzing the normalized spacing between each lens, along with the optimized shroud designs described in **Fig. 5**, similar observations can be made about the vortex interactions behind a wind lens.

Khamlaj and Rumpfkeil [8] established a high-fidelity CFD framework to develop a high power, low drag wind lens which was optimized for a wind speed of 8 m/s. The optimized lens profile was validated by comparing to the previously mentioned work done by Ohya and Karasudani [2]. The results of this work displayed that wind lens turbines could be improved upon using this optimized model. The results of the work yielded three lens profiles that had a higher coefficient of power than previously developed models which is used in the present study. Novotny and Gunasekaran [9] investigated the performance of the lens profile E that maximized the coefficient of power and minimized coefficient of drag through experimental measurements in UD-LSWT as well as full-scale field testing in both 1D and 2D configurations. Their work also included a disk testing (surrogate for the WLTs) in 3-D grid arrangement. Both the wind tunnel and the field test results showed significant increase in per unit power output when

installed in a grid configuration. As stated earlier, this increment is predominantly attributed to the aerodynamic interactions of the flow around the lenses. Hence, if the interactions can be understood in both 1D and 2D configurations, then grid spacings can be optimized.

To optimize the overall design of the lensed turbine grid system, the complex wake interactions between the aligned turbines are analyzed based on their proximity effects. In analyzing these interactions, the area of interest lies in the downstream space between each WLT, where the vortices begin to form (**Fig. 2**). Furthermore, studying the effects of both 1D and 2D arrangements will allow for the identification of the most efficient grid configuration that yields a higher power output. Utilizing both PIV (particle image velocimetry) and hotwire anemometry, the current study is aimed at quantifying the extent of wake interactions and resultant effect on the WLTs in 1D grid patterns, with inferences made about how these interactions could be implemented in both 2D and 3D configurations.

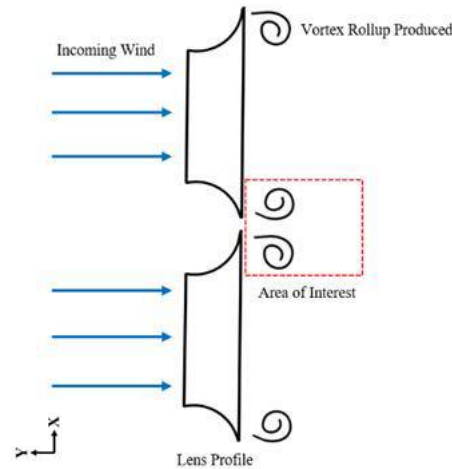


Figure 2: Schematic of Wake Interaction in Downstream Wake of Lens.

The result from two lensed turbines arranged in 1D configuration done by Novotny and Gunasekaran [9] is shown in **Fig. 3**. Here, the left turbine normal to the flow is displayed in grey and the right turbine normal to the flow is displayed in gold. It should be noted that both turbines are rotating in clockwise direction with respect to the freestream. The red line represents the system average for the two turbines and the blue line represents the results of the disk testing. The wind lenses displayed a peak increase of 16% at an x/d of 0.1 when compared to a single lensed turbine. However, it can also be seen that the right turbine outperforms the left turbine. This bias is hypothesized due to the swirl of the turbine and due to the bias in the wake direction driven by wake interactions which will be investigated in this research.

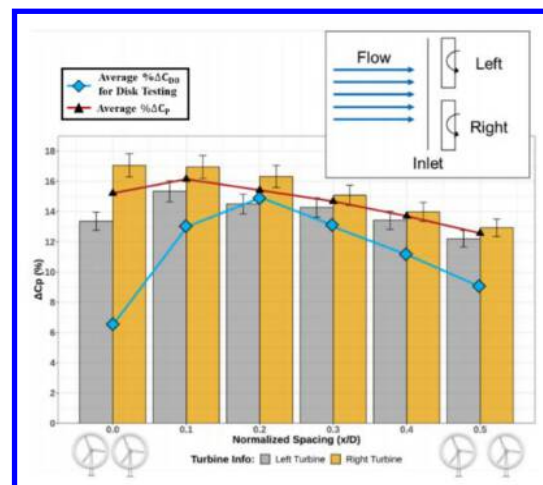


Figure 3: Percent Change in Coefficient of Power for 1D WLT Design.

III. Selection of Optimal Shroud Profile

The optimized design established by Khamlaj and Rumpfkeil [8] provided optimized lens designs that resulted in a high coefficient of pressure value (C_p of 1.003), with a lower drag coefficient (C_d of 0.876). This lens design, labeled profile E, was described using the design parameters identified in **Fig. 4**. Specific to this profile, the key parameters optimized were the normalized nozzle length, L_N , which had a value of 0.176, and the brim height, h , having a value of 0.161. The other design parameters specific to this profile are as follows: diffuser length (L_D) of 0.193, nozzle height (R_N) of 0.577, and diffuser height (R_D) of 0.616. By implementing this optimized design, both initial and future testing can be conducted using various scale-versions of this lens profile, both in single-plane and multi-plane experiments. Additionally, the two other optimized shroud designs (profiles C and D) proposed by Khamlaj and Rumpfkeil [8] were experimentally investigated to determine the optimal shroud design specific to the observable flow conditions. The description of these optimized shrouds along with their corresponding dimensions and relevant flow characteristics can be seen in **Table 1**, along with the profile designs in **Fig. 5**.

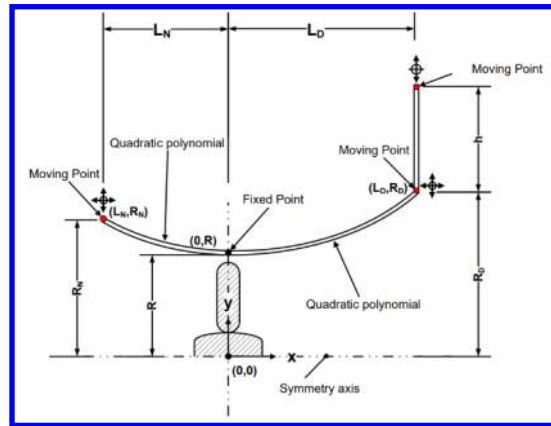


Figure 4: Design Variables for Parameterization in Shroud Optimization.

Table 1: Comparison between Optimized Profiles C, D, and E.

	L_D	h	R_N	L_N	R_D	C_p	C_d
Profile C	0.193	0.084	0.573	0.143	0.613	0.896	0.664
Profile D	0.193	0.198	0.577	0.169	0.616	1.056	1.055
Profile E	0.193	0.161	0.577	0.176	0.616	1.003	0.876

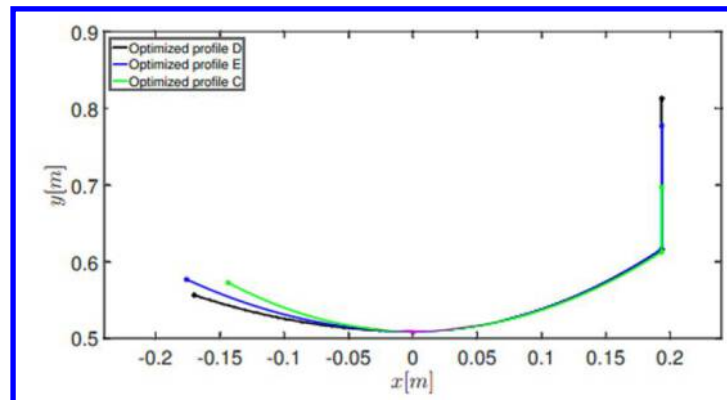


Figure 5: Optimized Lens Profiles (C, D, and E) which yield higher C_p and lower C_d .

IV. Experimental Setup

A. Wind Tunnel

All experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT). The UD-LSWT has a contraction ratio of 16:1 with 6 anti-turbulence screens. The freestream turbulence intensity is below 0.1% at 15 m/s measured by hotwire anemometer. The test section velocity varies from 3 m/s to 40 m/s with four interchangeable test sections. The tunnel also contains a downstream shuttering system that varies the freestream velocity profile up to 5 Hz and 50% of the velocity amplitude.

All the experiments were done in the open jet configuration. The test section inlet measures 76.2 cm x 76.2 cm (30 in x 30 in) and opens into a pressure sealed plenum. The effective test section length is 182 cm (72 in). A 112 cm x 112 cm (44 in x 44 in) collector returns the expanding air to the diffuser. The freestream velocity is measured using a Pitot tube connected to a TSI Micromanometer (Model 5825).

B. Linear Lens Profile Model

The aerodynamic interactions were investigated using the lens profiles which are linearly extruded to simulate two-dimensional flow without the effects of the curvature. The lens profile was scaled to produce an overall lens width, or chord, of 1 in (0.0254 m), with a material thickness of 1 mm. The lens span, or height of the profile, was set to 8 inches (0.2032 m), with a schematic of the preliminary setup seen in **Fig 6**. This model was secured to a custom mount, which allowed for the lenses to be adapted to change the various normalized spacing distances (x/d), as described in the previous study conducted by Novotny and Gunasekaran, which investigated only Profile E lens [9]. Specific to this experiment, five normalized spacing parameters (x/d) from 0.1 to 0.5 were investigated at a freestream velocity of 10 m/s, for each optimized profile described above (C, D, and E). The test matrix used for both experimental methods can be seen in **Table 2**.

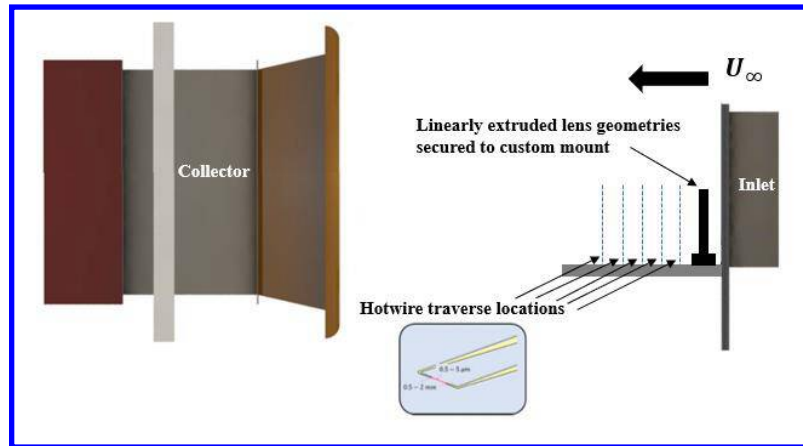


Figure 6: Schematic of lens profile testing setup with implemented design (hotwire experiment).

Table 2: Test matrix used for experiments.

Lens Profile	Lens Spacing (x/d)	Hotwire Spacing (x/c)	Hotwire Spacing (y/c)	Freestream Velocity (m/s)
Profile C	0.1, 0.2, 0.3, 0.4, 0.5	-4 to 4	1, 2, 3, 4, 5, 6	10
Profile D				
Profile E				

C. PIV Setup

The PIV experiment was conducted using a Vicount smoke seeder with glycerin oil, a 200 mJ/pulse Nd:YAG frequency doubled laser (Quantel Twins CFR 300) and an Imperx B2021 camera with 100 mm lens. The time delay between the laser pulses was set to obtain 8 to 10-pixel particle displacement in the wake of the lenses. The time delay

which produced the desired particle displacement was 94 microseconds (0.000094 s). A Powell lens was used in series to open the laser beam into a sheet. The laser and camera were triggered simultaneously using a Quantum composer pulse generator. In each test case, 1000 image pairs were obtained and processed using ISSI Digital Particle Image Velocimetry (DPIV) software. Two iterations were performed in DPIV processing with 64-pixel interrogation windows in the first iteration and 32-pixel interrogation windows in the second iteration. The PIV setup is shown in **Fig. 7**. The FOV was placed behind the two lenses approximately 1 in. downstream, as shown in **Fig. 7**. The size of the FOV was found to be 161.5 mm x 161.5 mm (6.36 in. x 6.36 in.), with a spatial resolution of 12.83 pix/mm (12830 pix/m).

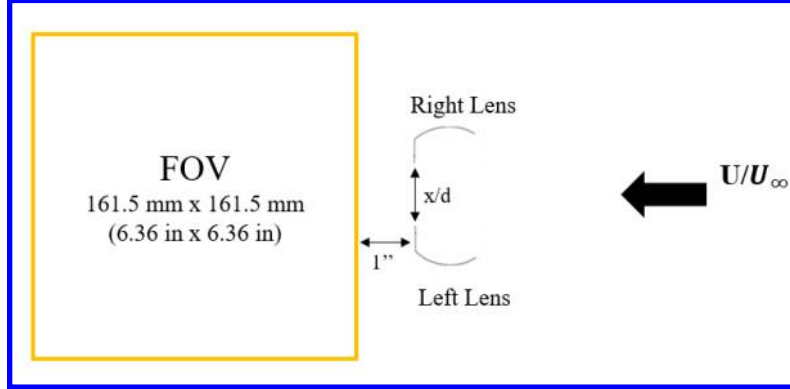


Figure 7: PIV setup for data collection of single-plane, empty lens profile.

D. Hotwire Anemometry Testing

The hotwire experiments were performed using a Dantec hotwire probe (type 55), with an overheat ratio of 0.45 for a wire temperature no greater than 300° C. The hotwire was placed at various downstream locations from the lens models (1-6 y/c.) and traversed laterally at ¼ in. increments (0.25 x/c) to collect data of the flow field downstream of the models. The hotwire locations are normalized by the chord (c) of each lens, which was 1 in. throughout. The traverse used was a UniSlide motor-driven assembly (model MA40), powered by a Velmex VXM-1 Stepper Motor. Data was collected using the MATLAB Data Acquisition Toolbox (DAQ). The flow field for each spacing parameter was varied laterally for each spacing parameter (x/d 0.1-0.5) and traversed downstream at the same y/c distances of 1 for each case. The downstream hotwire locations are shown in **Fig. 8**.

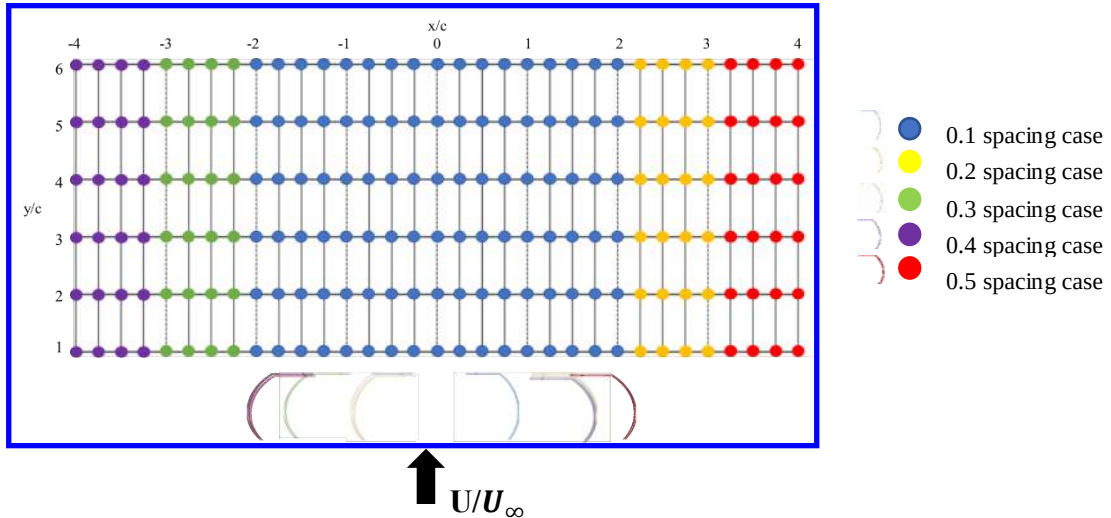


Figure 8: Hotwire locations with respect to the lenses, for each spacing case.

V. Results

A. PIV Results

The results from PIV are shown in this section for the different x/d spacing cases for all three lens profiles. The mean velocity contours shown in **Fig. 9** illustrate the mean interaction between the wake of the two identical lens profiles at different x/d spacings. At a x/d of 0.1, the lens profile D has a larger velocity deficit when compared to both lens profiles C and E. This trend is continued in all the other x/d cases. However, several distinctions in the wake behavior can be elucidated from **Fig. 9**. All lens profiles start with two distinct and independent wakes at 0.5 x/d . The momentum deficit magnitude is directly related to the drag coefficient of each individual lenses. At a distance of 0.4 x/d slight changes in the momentum deficit can be observed for Profile D and E with the right (with respect to the freestream) lens showing a slightly higher momentum deficit than the left lens. Reducing the spacing even further to 0.3 x/d , there is a drastic change in the magnitude of momentum deficit for Profile D by almost 20% while the magnitude of momentum deficit for Profile C and E remains the same. Strong deviations can be seen between the lens profiles at 0.2 x/d spacing. At this spacing, the wakes of profiles D and E seem to be merged, with the wake from Profile C seeming to only partially merge, and much further downstream. Distinct wakes emanate from each lens independently but within 3 x/c , the two wakes merge into one. At 0.1 x/d , all lens profiles exhibit only one wake indicating the wakes are completely merged. However, the location of the merged wake with respect to the lenses are different. The merged wake location in Profile C and D seems to occur in the centerline of the FOV where the two lenses meet. However, Profile E shows a highly skewed wake location where the merged wake is seen behind the left (with respect to the freestream) lens profile. There is almost no indication of the presence of the right lens in Profile E case at 0.1 x/d . As seen in **Fig. 3**, the highest increment in C_p is around x/d of 0.2, beyond which the increment in C_p decreases. Therefore, a strong correlation can be made between the C_p increment observed at x/d of 0.2 and the mean aerodynamic interaction of the wake.

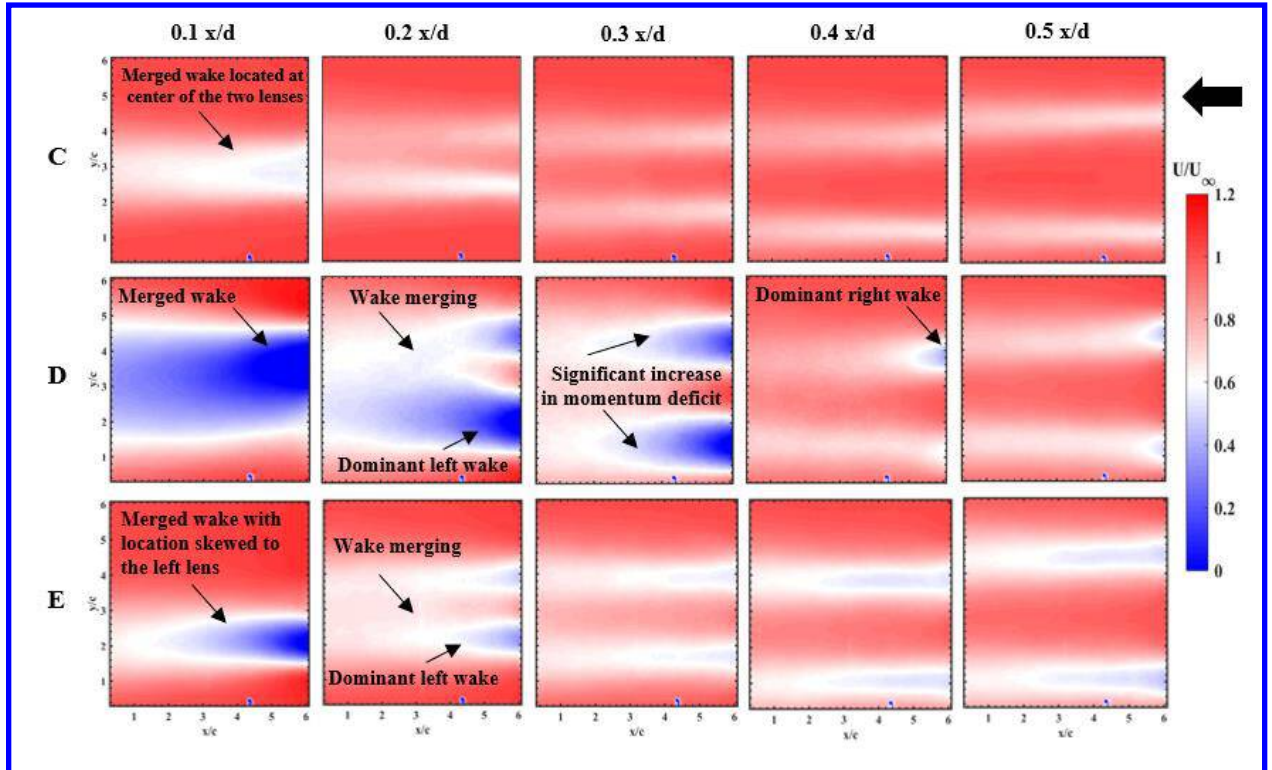


Figure 9: U/U_∞ for each profile (C, D, and E), along with each spacing parameter (x/d).

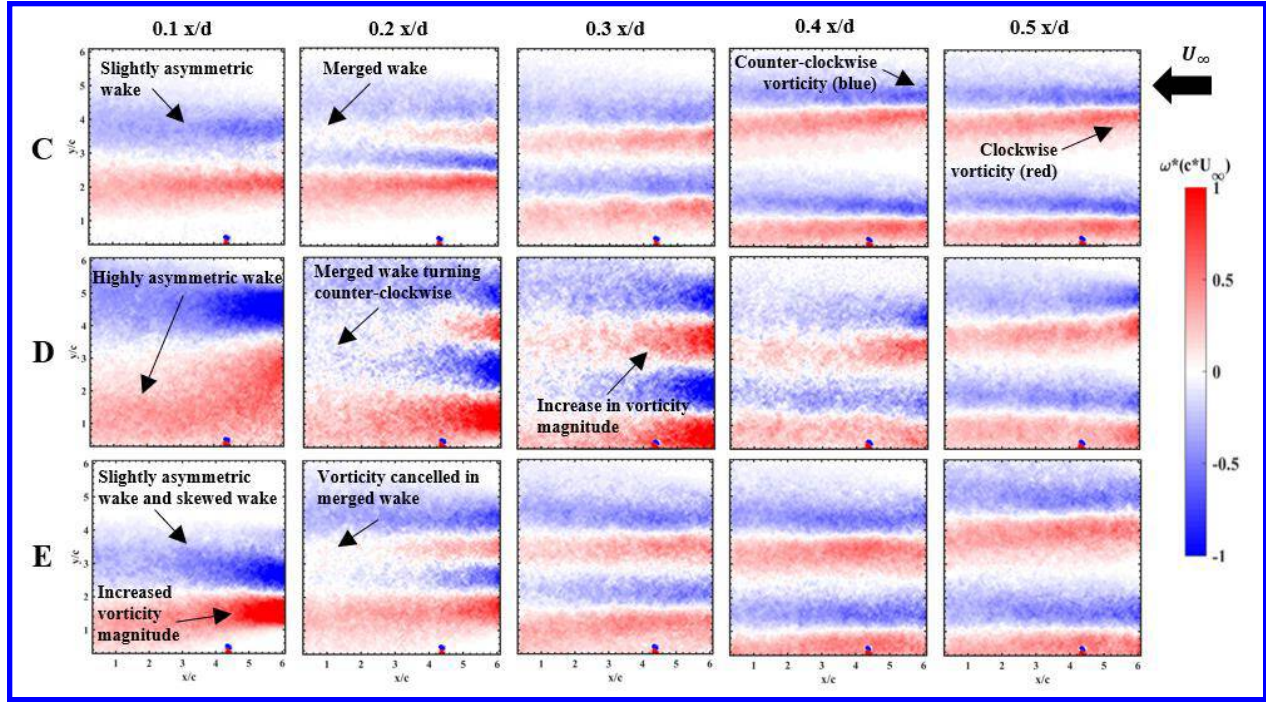


Figure 10: Vorticity for each profile (C, D, and E), along with each spacing parameter (x/d).

Similar trends can be observed in the mean vorticity contours shown in **Fig. 10** with additional insight into the wake interaction. The two bands of alternate vorticity can be seen from each lens for all lens profiles. The red vorticity region indicates clockwise rotation, and the blue vorticity region indicates counter-clockwise rotation. At 0.5 x/d , the wakes of the Profile C lens pairs seems farther apart than Profile D and E lens pairs. This could be since Profile C has a lower drag coefficient and thus a smaller momentum deficit and a thinner wake when compared to the other two lens profiles. No tangible differences are seen in the vorticity magnitude for the two lens pairs in each profile case. Even though one of the wakes seems to dominate in the mean U velocity cases, the vorticity magnitude remains almost equal for each lens pairs. However, a significant jump in vorticity magnitude can be seen for Profile D at 0.3 x/d which also corresponds to significant increase in momentum deficit seen in **Fig. 10**. The merging of the wake can be seen for all lens profiles at 0.2 x/d . While the merged wake turns to have a more counter-clockwise rotation (blue) for Profile D case, the mid-vorticity between the two lens pairs cancel each other in the case of Profile E. This could be due to the fact that the upper surface vorticity (blue) of the left lens has a slightly stronger magnitude than the lower surface vorticity (red) of the right lens that the resultant wake has a net counter-clockwise rotation after merging. In case of Profile E, these two vorticity bands from the lens pairs cancel each other resulting in an almost irrotational flow. This could be one of the reasons why the merged wake ends up skewed to the left lens instead of occurring at the center of the two lenses. The symmetry of the wake can also be visualized in **Fig. 10** for all lens profile cases at each x/d . All lens profiles seem to exhibit a symmetric wake profile at 0.5 x/d and maintain symmetry until merging at 0.2 x/d . After merging however, the wake profiles tend to be slightly asymmetric in the case of Profile C and E whereas high asymmetry in the wake can be seen for Profile D. The vorticity contours clearly show that the dynamics of wake merging is strongly dependent on lens geometry and offers insight into the skewness and asymmetry of the wake after merging.

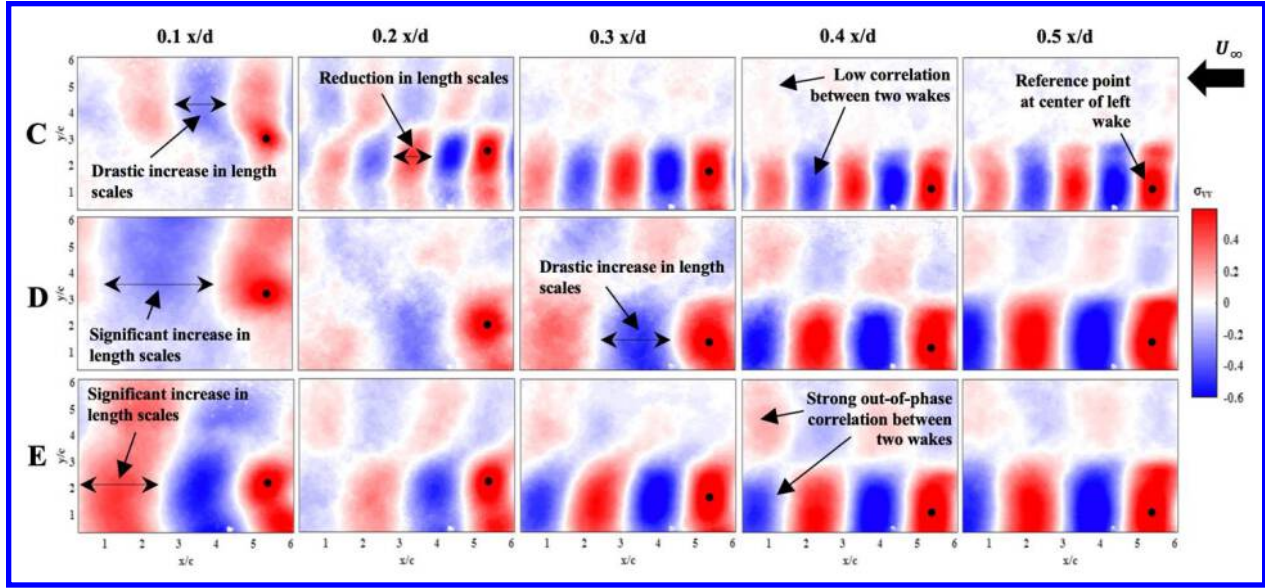


Figure 11: Two-point correlation for each profile (C, D, and E), along with each spacing parameter (x/d).

It has been well known that the vortex shedding frequency and turbulent length scales contributes to turbulent-induced pressure fluctuations, sound generation, and structural vibrations [10]. The effect of lens proximities in the wake and the vortex shedding frequency and the corresponding length scales can be determined by comparing the coherent structures in the wake for different lens profiles and different proximities. The coherent structures can be determined by performing two-point correlation of fluctuating velocities in the wake. Bendat and Piersol [11] defined the two-point correlation as

$$\rho_{u_i u_j}(X_1, X_2, \tau) = \frac{[u'_i(X_1, t) * u'_j(X_2, t + \tau)]}{\sqrt{u'^2_i(X_1)} \sqrt{u'^2_j(X_2)}} \quad (1)$$

X_1 and X_2 are two spatial locations in the PIV field of view, τ is the time delay (which is chosen to be zero for the results shown below), and u' represents the fluctuating velocities in i and j direction. The two-point correlations for the V velocity component are shown in **Fig. 11** for all lens profiles and proximities. The reference point X_1 was chosen to be at the center of the wake of the left (with respect to the freestream) lens. The alternate band of correlations can be seen for all cases in **Fig. 11** which approximates the presence of coherent structures that are shed from the lenses and their corresponding length scales. At 0.5 x/d , it can be seen that the Profile C exhibits a lower length scale, higher frequency coherent structures when compared to Profile D and E. While reasonable out-of-phase correlations persists between wakes of the left and the right lenses for Profile D and E, the correlations between the left and right wakes are less pronounced in the case of Profile C. As the distance between the lenses are decreased at 0.3 x/d , Profile D exhibits a drastic increase in the length scales which has a strong influence in the mean velocity and vorticity magnitude as seen in **Fig. 9** and **Fig. 10**. No such changes are seen for Profiles C and E at 0.3 x/d . At 0.2 x/d , all lens profiles exhibit a loss in the strength of the correlations when compared to higher x/d cases. This could be due to the wakes merging into each other. Apart from changes in correlation magnitude, Profile C exhibits a decrease in length scales as 0.2 x/d when compared to 0.3 x/d . No such changes in the length scales are seen in Profile D and E at 0.2 x/d . However, the merged wake on all lens profiles for 0.1 x/d shows a drastic increase in the length scales, with Profile D exhibiting a significant increase and Profile E exhibiting the least significant increase. The correlations show that the merged wake of Profile C is slightly skewed to the right lens instead of the left as seen in Profile E. These correlations provide strong evidence to fundamental changes in the wake signature as profile geometry and proximity changes. Based on these observations, significant conclusions can be drawn not only about the downstream properties in the area of interest between the lens profiles, but perhaps most importantly the increment in power identified by Novotny and Gunasekaran [9].

B. Hotwire Anemometry Results

The temporally resolved hot-wire data was used to determine the turbulence intensity and the spectral content of the wake through Fast Fourier transform. The turbulence intensity is calculated by Equation 2.

$$TI (\%) = \frac{\sqrt{u'^2}}{U_\infty} * 100 \quad (2)$$

where u' is the fluctuating velocity in the wake. The TI calculated at each grid location shown in **Fig. 8** is plotted as a function of x/c and y/c in **Fig. 12**. The bimodal nature of the TI profiles at higher lens distances shows the two independent wake signatures where higher TI magnitude is seen directly behind the lenses and gradually decreasing with increase in downstream distance. While the decrement in TI in the wake of the two Profile C and E lenses show similar decay, the wakes of Profile D lenses show widely different behavior where the right wake decays significantly when compared to the left wake, even at a greater lens spacing. This could be due to the higher wake width created by the Profile D lens as seen in the PIV results. The decay of TI in the right lens when compared to the left becomes highly evident in Profile E at $0.3 x/d$ indicating that the wake merge takes place at downstream distance rather than directly behind the lenses. The profile C case at $0.3 x/d$ however show two stratified wake signatures without any indication of merging. At $0.2 x/d$ the wakes of all lens profiles are merged, and it is at this stage where the skewness of the wake can be seen. Significant skewness in Profile D and Profile E cases can be observed at $0.2 x/d$ lens spacing where the wake is skewed to the left lens. Partial skewness towards the left lens can also be observed for Profile C case as well. At $0.1 x/d$, the bimodal nature of the TI profile disappears as the wakes merge into one wake with a high TI magnitude. However, the TI decay rate of the merged wake is significantly less when compared to the independent wakes of the lenses at $0.5 x/d$. The skewness in the wake could be the indicator of why two wind lens turbines placed at close proximities show slightly different performance in power output. **Fig. 12** along with the PIV results provides insight into the disparity in the power output between lens pairs can be observed clearly in Novotny and Gunasekaran [9] and shown in **Fig. 3**.

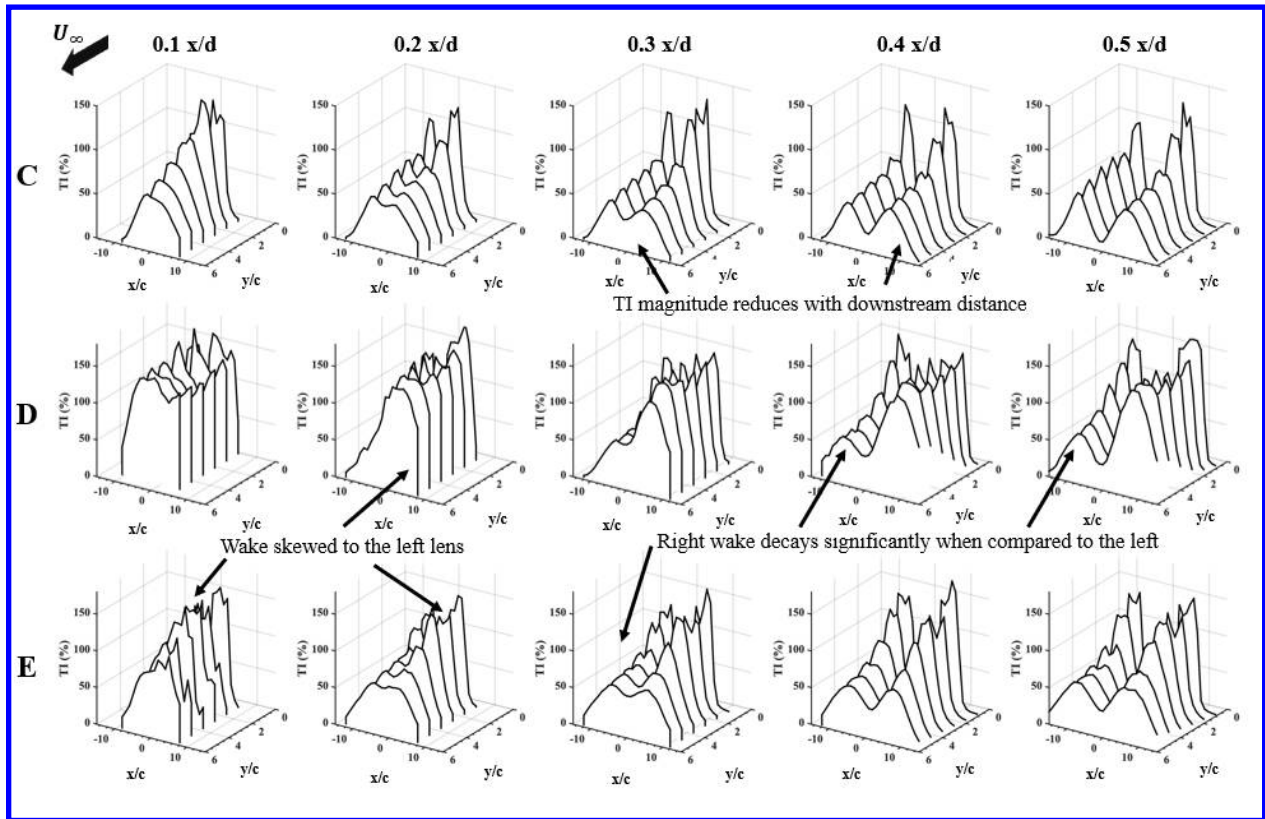


Figure 12: Wake turbulence intensity results from hotwire data at different downstream locations in the wake of lens profiles

While the wake skewness could explain why only one lens turbine in lens turbine pair outperforms the other, it doesn't quite explain the significant increase in the overall net power output seen in the wind lens pairs. The spectral content of the wake is investigated to determine a plausible cause for the net power output. The fast Fourier transform (FFT) was performed to the velocity signals obtained from hot-wire measurements at each grid point locations shown in **Fig. 8**. The spectral content in the wake at 3 y/c downstream distance is shown in **Fig. 13** at various x/c cases tested. At 0.5 x/d lens spacing, all lens profile cases show a distinct wake signature for the lens pairs. The frequency spectrum at the wake of the lenses is scattered with oscillations in frequencies when compared to the freestream. The peak frequencies in all the independent wakes of each lens pairs are around 100 Hz, which corresponds to a Strouhal number of roughly 0.2 depending on the chord of the lens. It is well known that the Strouhal number of 0.2 is the most common shedding frequency experienced in most bluff bodies. Even at a lens spacing of 0.5 x/d and 0.3 x/d, the dominant wake frequency of the left wake is lower than that of the right lens indicating that even though the wakes may seem independent, the effect of the neighboring wake is felt in its counterpart long before the wakes are merged. At 0.2 x/d, significant changes to the frequency spectrum can be seen. The peak frequency strength around 100 Hz seems to diminish, with Profile D showing no peaks at 100 Hz. The strengths of the lower frequencies however seem to increase when compared to the higher x/d cases. At 0.1 x/d, the peak frequencies around 100 Hz completely vanish for all the lens profile cases and the prominent frequencies shifted to the lower end of the spectrum. This phenomenon seems to coincide with the two-point correlation results seen in **Fig. 11** from the PIV data. At 0.5 x/d, the shorter length scales correspond to higher frequencies and 0.1 x/d, the large length scales correspond to lower frequencies. This reduction in peak frequency phenomenon explains clearly why the performance of the turbines are enhanced at close proximities. Due to the larger shedding frequency at larger spacings between the lenses, the “resting” time of the vortex behind the lenses are shorter. As the lenses are brought to close proximities, the merging of the wake disrupts this high rate of vortex shedding and lowers the shedding rate to lower frequencies such that the resting time of the vortex behind the lens is comparatively longer. A longer resting time of the vortex behind the lens corresponds to a more consistent lower pressure that draws air at a higher velocity at a steady rate compared to high vortex shedding frequency at larger separation distances.

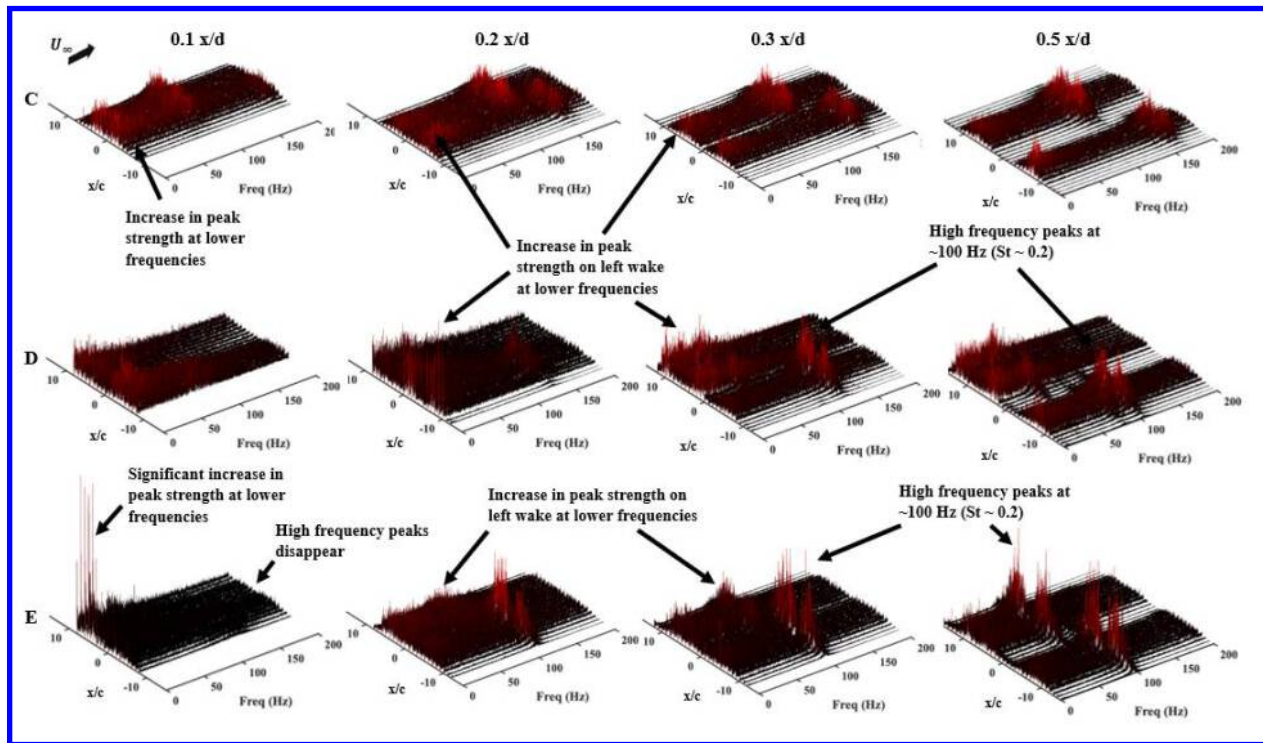


Figure 13: FFT plots highlighting dominant frequencies at 3 y/c downstream the lenses.

VI. Conclusions

The fundamental flow physics responsible for the increased net power output from a pair of lens turbines placed at close proximity is revealed through PIV and hotwire investigations. As the two lenses are brought to close proximities to each other, the following happens:

1. The two wakes sense the presence of each other such that the peak shedding frequency for one of the wakes is lowered when compared to the other. This causes changes in the mean flow parameters such as the mean velocity and vorticity.
2. Further bringing the lenses closer to each other signifies the onset of wake merging that causes significant changes in the vortex shedding frequency to one of the wakes. This results in that wake having a higher momentum deficit and a higher turbulence intensity.
3. The choice of the direction of the skewness in the wake is hypothesized to be due to the extent of cancellation on the inboard vorticity magnitude from the two wakes. If the vorticity is perfectly cancelled between the wakes, little to no wake skewness is observed. The direction of skewness of the merged wake causes one of the turbines in the lens turbine pair to outperform the other as shown in the previous study.
4. When the lenses are brought even closer, the high frequency shedding disappears, and the lower frequency shedding starts to dominate. This results in a longer resting time of the vortex behind the lenses that results in a more stable low-pressure zone, which draws in air more consistently at higher speeds than when the lenses are spaced at greater distances.

Acknowledgements

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References

- [1] Unlocking Our Nation's Wind Potential.” *Department of Energy*, 19 May 2015, www.energy.gov/eere/articles/unlocking-our-nation-s-wind-potential.
- [2] Ohya, Y., & Karasudani, T. (2010). A shrouded wind turbine generating high output power with wind-lens technology. *Energies*, 3(4), 634-649.
- [3] Ohya Y, Miyazaki J, Göltenbott U, Watanabe K. Power Augmentation of Shrouded Wind Turbines in a Multirotor System. *ASME. J. Energy Resour. Technol.* 2017
- [4] Ohya, Y. (2019). Multi-Rotor Systems Using Five Ducted Wind Turbines for Power Output Increase (Multi Lens Turbine). In *AIAA Scitech 2019 Forum* (p. 1296).
- [5] Ohya, Y. (2014). Bluff body flow and vortex—its application to wind turbines. *Fluid Dynamics Research*, 46(6), 061423.
- [6] Zhou, H., Zhang, H. J., & Yiu, M. W. (2002). The turbulent wake of two side-by-side circular cylinders. *Journal of Fluid Mechanics*. Vol. 458, pp 303-332.
- [7] Teimourian, Amir, Yazdi, Sina G., & Hacisevki, Hasan. (2018). Vortex Shedding: A Review on Flat Plate. *Fluid Dynamics*. Vol. 53, No. 2, pp 212-221.
- [8] Khamlaj, T. A., & Rumpfkeil, M. (2018). Optimization Study of Shrouded Horizontal Axis Wind Turbine. In *2018 Wind Energy Symposium* (p. 0996).

- [9] Novotny, Neal, and Sidaard Gunasekaran. "Performance and Proximity Investigations on Small Scale Lensed Turbines." In AIAA Scitech 2020 Forum, p. 1242. 2020.
- [10] Gunasekaran, Sidaard, and Daniel Curry. "On the Wake Properties of Segmented Trailing Edge Extensions." *Aerospace* 5, no. 3 (2018): 89.
- [11] Bendat, Julius S., and Allan G. Piersol. *Random data: analysis and measurement procedures*. Vol. 729. John Wiley & Sons, 2011.